

A Comparative Analysis of Machine Learning Algorithms for Student Performance Prediction

Md Sakib Sadman Badhon

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

sakib.sadman.badhon@g.bracu.ac.bd

Siam Ferdous

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

siam.ferdous@g.bracu.ac.bd

Abstract—Education is a basic human right and people all over the world take some form of education in their lifetime. Among them, most people get their education from an institution. But as the population grows, good institutions do not build that fast, so the competition to get into a good institution rises. Because of that, institutions face numerous challenges in identifying the appropriate candidates to enroll. Not to mention, after getting enrolled in an institute, many students fail to reach their full potential because they receive generalized care. This study explores the seven machine learning (ML) algorithms, including K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Logistic Regression, Support Vector Machine(SVM), Naive Bayes and Gradient Boosting , to ease the candidate selection processes and find those students who need more attention. These seven ML algorithms are evaluated in an identical environment to determine which algorithm provides better accuracy and efficiency in finding the condition of a student. These findings will help the institutions make informed decisions while consuming less time and resources.

Index Terms—Student, Performance, Algorithm, Machine, Learning, Prediction

I. INTRODUCTION

Ensuring quality education is crucial for the development of any country. When it comes to student's academic achievements, it is a major concern for education institutes as it impacts students' future careers and Institution's reputation [1]. Throughout the previous few years, machine learning has been seen as a promising tool in the education sector [2]. This area keeps building huge data reserves from different systems like admissions, academic data and e-learning platforms. Machine learning algorithms, that are designed to use this data to make predictions, function as a model approach for the future of predicting student academic performance. For example, teachers can forecast the grades and identify the students who require assistance at an early stage and then offer student-centered support that will help the students reach the highest standard. Furthermore, it can help educational institutions to enhance their teaching skills and overall educational system. While numerous studies have explored the application of machine learning for student performance prediction, a comprehensive evaluation of multiple algorithms across various educational contexts is lacking. Previous works mostly focused on specific machine learning algorithms failing to provide a compre-

hensive awareness of the predicting potential capabilities of various ML approaches.

This paper aims to fill this research gap by conducting a comparative analysis of the accuracy of various Machine learning models, including Decision Tree, Random Forests, Gradient Boosting and four other models for predicting student academic performance. With the assistance of multi-level education data which is composed of student's demography, educational histories and learning patterns, this will help us to learn which model is best suited for improving educational outcomes as well as optimizing time efficiency for different use cases.

II. LITERATURE REVIEW

As this is an old field, many rigorous studies were conducted to predict student performance with machine learning algorithms. The authors of [3] tried to use a multiple Linear Regression model to identify factors contributing to students' academic achievements. In their study they used Internal factors, external factors and combined factors of internal and external factors. They concluded the research by stating that ML learning algorithms would have provided more accurate results than the multiple Linear Regression model. Following their findings lets look into some research which includes ML model to predict the performance of a student.

In [4], the authors selected 300 bachelor's degree student from five different colleges and created a dataset containing 17 features, including students grades, living standards, student activities, family income and family status. They used Naive Bayes, Decision Tree and Neural Networks to predict student's course success and found that Naive Bayes performed better than the other two. For training and testing purpose they used the KNN model and implemented Linear Regression.

In paper [5], the authors choose seven different ML algorithms, including Decision Tree, Naive Bayes, Random Forest, KNN, SVM, Artificial Neural Networks (ANNs) and Logistic Regression, on three different types of task-oriented binary and ternary classification educational datasets. This dataset contains information of student admission, engineering placement information and student performance information. They found out that Random Forest has achieved greater accuracy among all the models and KNN shows the worst

performance. Also, They did not recommend Naive Bayes for any of the datasets.

The authors of [6] tried to predict a student's best subject and learning outcome by using KNN, Gaussian Naive Bayes (GNB), SVM, Genetic Bee Algorithm (GBC), voting classifiers, Random Forest Classifier (RFC) and Decision Tree with and without Principal component analysis (PCA). After conducting a thorough analysis, they found out that GBC, GNB, Decision Tree and Support Vector Classifier (SVC) are the best in their respective areas, achieving accuracy of near 70%. Which is not the best among all the other study.

In the research paper [7], the authors tried to predict student performance using five machine-learning algorithms. They have used two datasets, one for prediction and one for classification. They have used Back-Propagation (BP), Support Vector Regression (SVR) and Long Short-Term Memory (LSTM) to predict student performances. Moreover for classification dataset, they have used BP, Gradient Boosting Classifier (GBC) and Support Vector Machine(SVM) to get the most accurate result. According to the experiment, BP scored 87.78% accuracy score in the 30% testing ratio, which is greater than other machine learning algorithms they have used. In the conclusion, They have talked about the shortage of the proper dataset to test the algorithms in this filed.

[8] touches on the surface of neural networks by introducing Artificial Neural Networks (ANN). Their dataset consists of 200,000 records of Colombian University students who sat for the SABER PRO test in 2016. This data contains students personal information and academic information. The authors conducted the test on Neuro-Solutions 7.1, which is a neural network package. They randomly initiated weight and trained the model 30 times (200 epochs). According to their experiment, ANN didn't perform so well compared to other ML models like Gradient Boosting and random forest.

III. METHODOLOGY

The author of this paper sourced the dataset from Kaggle. The name of the dataset is "Student Performance". The dataset consists of two Comma Separated Value (CSV) files, namely "Maths.csv" and "Portuguese.csv," both of which contain the same number of features and target values. In order to produce a more accurate value and increase the number of variables, the datasets were merged. One important thing to mention is that this dataset does not contain any personally identifiable information; rather, the data holds non-identifiable personal and educational aspects of a student, ensuring full anonymity.

Initially, the dataset has 34 features and 1048 columns. Among them, there were 3 different columns, G1, G2 and G3, which represent the first period grade, second period grade and final grade, respectively. Here, the target feature is the final grade (G3). The authors of this paper applied the binning method to the target column, categorizing the grades as follows: "Low" for scores ranging from 0 to 6.67 (labeled as 1), "Medium" for scores ranging from 6.67 to 13.33 (labeled as 2) and "High" for scores ranging from 13.33 to 20 (labeled as 3). This dataset contains 1048 values; in between those values,

there were no missing or unusual values, so imputation was not required for this study.

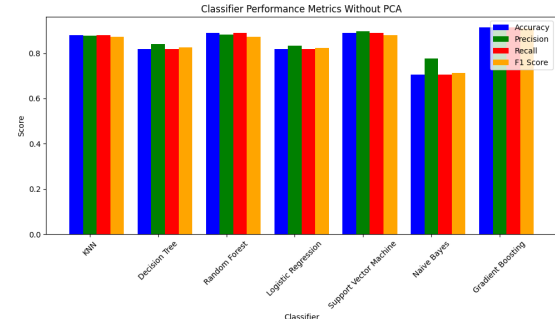
The dataset consists of 17 categorical variables and 14 numerical variables. As machine learning models cannot handle these categorical values, encoding was used. There are two popular encoding methods, which are one hot encoding and label encoding. One hot encoding is infamous for increasing the dimension of the dataset after encoding, but as the dimension of the selected dataset is already quite large, choosing one hot encoding might reduce the model training performance significantly. That is why the author of this paper chose label encoding as their primary encoder. Label encoding works best when there are fewer unique values in the features and the selected dataset fits this condition properly. Moreover, the label encoding will not increase the dimension of the dataset, which will keep the training process of the models as efficient as before.

Furthermore, after conducting a feature correlation analysis, it can be found that almost all the features are not significantly correlated with each other. But the G1 and G2 features are highly correlated with each other, scoring 85% in correlation. This high correlation means that keeping both features in the dataset will not add any value to the model, as 85% of their values are similar to each other. For this study, the authors decided to drop those columns that have more than 70% similarities. So they dropped the G2 feature from the dataset during the model training and trained the model with other features.

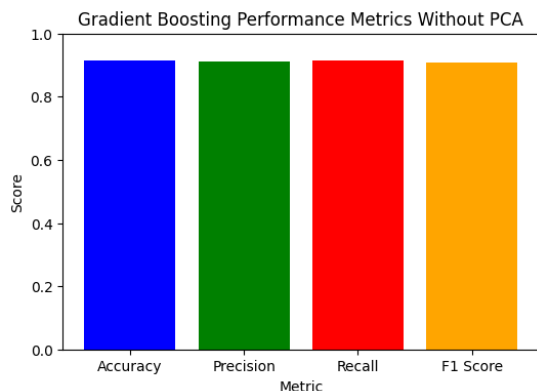
This dataset is divided into training and testing sets in an 80-20 ratio. Here, the training set has 838 values and 210 values on the testing set. To continue further analysis of this dataset, the authors used seven machine learning algorithms from the scikit-learn library. The accuracy, precision, recall and f1-score of each algorithm were assessed both with and without the application of principal component analysis (PCA). These performance metrics will provide an insight into how well a model is capturing the underlying complex relationships in this dataset. To ensure fairness and consistency in evaluating the algorithms' performance, all of the algorithms are tested on the same set of hardware on Google Colab.

IV. RESULTS AND ANALYSIS

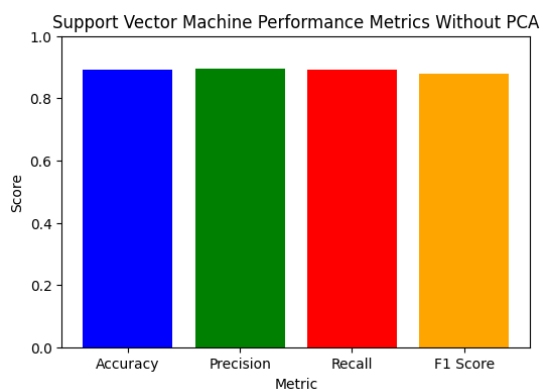
To analyze the effectiveness of the seven machine learning algorithms to predict student performance more accurately, four different performance metrics—accuracy, precision, recall and f1-score were used.



Overall, from these findings, it can be noticed that Gradient Boosting has the best ability to predict student performance without PCA. It makes predictions through the use of an ensemble learning technique that combines numerous trees to make predictions. By following a sequence and fitting new models residually on previous models, it can give more refined predictive accuracy. Which helps this model give better results than other models. Moreover, it works well with non-linear patterns between features and the target variable; in this case, the Gradient Boosting model is more effective in capturing such relations. showed its strengths with an accuracy of 91%, precision of 91%, recall of 91% and f1 score of 91%.

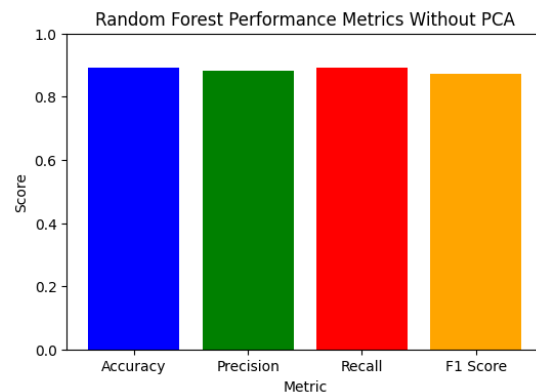


Support Vector Machines (SVM) focus on finding the hyperplane that maximizes the margin between the classes, which helps it effectively separate the data points and classify them into different categories. In our case, it indicates its capability to classify instances and capture true positives effectively and correctly. However, its achieved accuracy of 89%, recall of 89%, precision of 90% and F1 score of 88% fell slightly short of Gradient Boosting . It may be due to its struggle to reduce false positives.

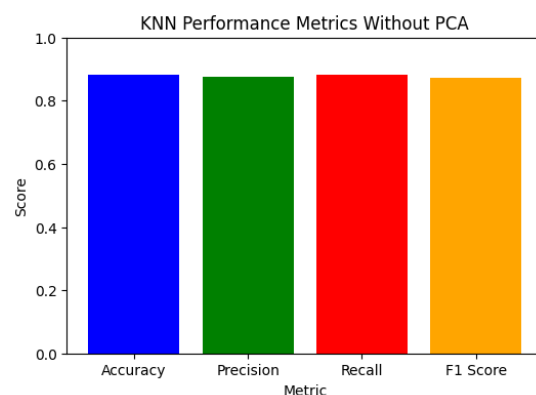


Similarly, Random Forests showed its accuracy: 89%, precision 88% recall 89% and a f1 score of 87%. It also uses ensemble learning methods like Gradient Boosting. But it has a different mechanism. Random forests add trees independently. After that, it derives the predictions

by majority vote for classifications. It helps to reduce the overfitting issue and make the model maintain its performance as the trees are less correlated to each other. On the other hand, Gradient Boosting adds Decision Trees one by one, creating and then correcting the errors one by one. As the trees correct the mistake made by the previous one, it gives a more optimal solution. It is more suitable for capturing complex relationships between features.

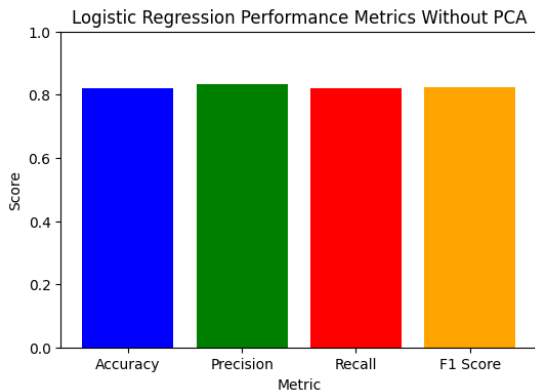


K-Nearest Neighbors (KNN) achieved accuracy 88%, recall, precision and an F1 score of 88%, 88% and 87%, respectively, without PCA. KNN is heavily based on a distance metric to determine the closeness between data points. The difference in distance metrics (Euclidean distance, Manhattan distance and cosine distance) may have an impact on the algorithm's efficiency. Different scales or unimportant features can lead to the wrong distance calculation, which can affect the classification accuracy. KNN performance suffers when dealing with high-dimensional data. When the number of dimensions increases, the Euclidean distance between data points may not correctly represent the structure of the data, which may imply lower efficiency and more complexity. Like the above model, it also has parameter sensitivity and imbalanced data can affect the accuracy.

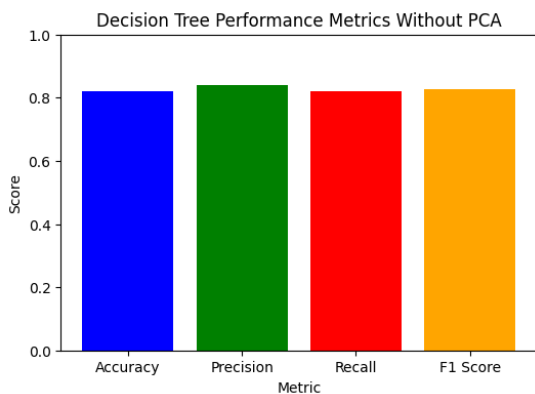


From the analysis, it can be noticed that Logistic Regression achieved an accuracy of 82%, precision of 83%, recall of 82% and F1 score of 82%. This model assumes that the

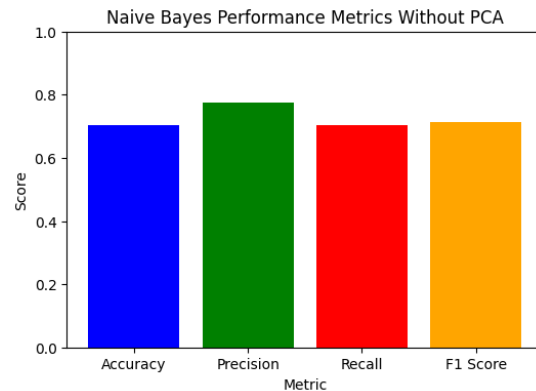
relationship between the independent variable and the respective dependent variable is a linear one. It may not analyze the complexity of the nonlinear relationships in the data set as effectively as other models. The number of features, outliers and distribution of the target variable may affect its performance.



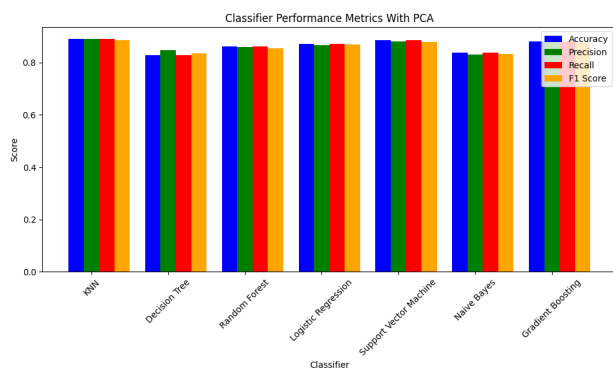
The Decision Tree achieved 82% accuracy, 84% precision, 82% recall and an F1 score of 83%. It can analyze non-linear relationships, but it has a tendency to overfit the data, especially when the depth of the tree and the number of nodes it contains are deep and complex. It is data-sensitive. Small variations in the training data led to the tree being built differently, which affected its performance. Also, trees are not globally optimized; they make local decisions on each node based on available data, which may not give the best overall result.



Without PCA, the Naive Bayes Classifier performed the least, achieving an accuracy of 70%, precision of 78%, recall of 70% and F1 score of 71%. This algorithm assumes all the features are independent, but in the selected data set, all the features are not independent, as can be seen in the heatmap. As a result, the classifier faces problems in modeling the conditional probabilities of features given the classes, resulting in poor performance.



The PCA enhancement of the classifiers led to better performance for KNN, Decision Tree, Logistic Regression (change in accuracy +5%) and Naive Bayes (change in accuracy +14%). PCA reduces the dimensionality by converting the data feature space by creating new orthogonal features that are called principal components and have zero correlation with each other. This simplification helps classifiers by cutting down the input space and handling the over fitting issue, especially in high-dimensional data. Which helps the accuracy of some models. However, this drop in interpretability and the dimensionality reduction can affect different classifiers with PCA. For this reason, random forest and gradient-boosting accuracy decreased. SVM performance was not impacted by PCA since SVM has the capacity to perform dimensionality reduction intrinsically by means of a hyperplane that divides different classes in the feature space.



It can also be noticed that a huge performance bump for Naive Bayes achieved accuracy, recall, precision and an F1 score of 84%, 83% and 83%, respectively, because PCA reduces the dimensionality of the feature space by capturing the most significant sources in the data set, eliminating the noise error and focusing on the most informative features that have enhanced Naive Bayes performance. As previously mentioned, Naive Bayes assumes feature independence, so PCA helps the models by solving multicollinearity by using a feature transformation, the technique that converts the input data into a new set of independent features with less correlation.

V. CONCLUSION

In conclusion, Gradient Boosting showed the best performance without PCA. Moreover, it can be noticed that PCA decreases the performance of Gradient Boosting and increases the performance of KNN, making it the best performer. The effectiveness of each model depends on the data set's properties and preprocessing techniques. PCA can improve the performance of some models that are unable to give accurate results due to the high dimensionality and noise in the data set. Therefore, it is important to carefully choose the characteristics of the data set and specify the problem before picking the right classifier.

Although it was not possible to carry out complete model tuning due to time limitations, some tuning methods like feature engineering, model ensemble and regularization may improve model accuracy and could be implemented in future studies. Furthermore, other models that are well-suited for non-linear data sets, such as XGBoost, LightGBM and CatBoost, could have been implemented in this research to recognize complex patterns in the data set and make the model more appropriate.

REFERENCES

- [1] A. M. Shahiri, W. Husain, and N. A. Rashid, "A review on predicting student's performance using data mining techniques," *Procedia Computer Science*, vol. 72, pp. 414–422, 2015, the Third Information Systems International Conference 2015. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1877050915036182>
- [2] A. Dutt, M. A. Ismail, and T. Herawan, "A systematic review on educational data mining," *IEEE Access*, vol. 5, pp. 15 991–16 005, 2017.
- [3] R. Sharma, S. K. Maurya, and K. Kishor, "Student performance prediction using technology of machine learning," 01 2022.
- [4] D. S. N. S. Ag Isha and S. R. Mohd. Hashim, "Application of multiple linear regression in identifying contributing factors in students' academic achievement," 12 2022.
- [5] Y. Chen and L. Zhai, "A comparative study on student performance prediction using machine learning," *Education and Information Technologies*, vol. 28, no. 9, pp. 12 039–12 057, Sep 2023. [Online]. Available: <https://doi.org/10.1007/s10639-023-11672-1>
- [6] T. Li, "Students' numeracy and literacy aptitude analysis and prediction using machine learning," *Journal of Computer and Communications*, vol. 10, pp. 90–103, 01 2022.
- [7] B. Sekeroglu, K. Dimililer, and K. Tuncal, "Student performance prediction and classification using machine learning algorithms," in *Proceedings of the 2019 8th International Conference on Educational and Information Technology*, ser. ICEIT 2019. New York, NY, USA: Association for Computing Machinery, 2019, p. 7–11. [Online]. Available: <https://doi.org/10.1145/3318396.3318419>
- [8] C. F. Rodríguez-Hernández, M. Musso, E. Kyndt, and E. Cascallar, "Artificial neural networks in academic performance prediction: Systematic implementation and predictor evaluation," *Computers and Education: Artificial Intelligence*, vol. 2, p. 100018, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666920X21000126>